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Instructor(s): *Mueller/Korytov*

PHYSICS DEPARTMENT

PHY 2049 | Exam 3 | December 10, 2019

Name (print): ______________________ Signature: ______________________

On my honor, I have neither given nor received unauthorized aid on this examination.

YOUR TEST NUMBER IS THE 5-DIGIT NUMBER AT THE TOP OF EACH PAGE.

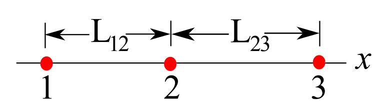
DIRECTIONS

(1) **Code your test number on your answer sheet (use 76–80 for the 5-digit number)**. Code your name on your answer sheet. **Darken circles completely (errors can occur if too light)**. Code your student number on your answer sheet.
(2) Print your name on this sheet and sign it also.
(3) Do all scratch work anywhere on this exam that you like. At the end of the test, this exam printout is to be turned in. No credit will be given without both answer sheet and printout with scratch work.
(4) Work the questions in any order. Incorrect answers are not taken into account in any way; you may guess at answers you don't know.
(5) If you think that none of the answers is correct, please choose the answer given that is closest to your answer.
(6) **Blacken the circle of your intended answer completely, using a number 2 pencil**. Do not make any stray marks or the answer sheet may not read properly. Completely erase all incorrect answers, or take a new answer sheet.
(7) As an aid to the examiner (and yourself), in case of poorly marked answer sheets, please circle your selected answer on the examination sheet. Please remember, however, that in the case of a disagreement, the answers on the bubble sheet count, NOT what you circle here. Good luck!!!

>>>>>>>>**WHEN YOU FINISH** <<<<<<<<
Hand in the answer sheet separately.

Constants			
$e = 1.6 \times 10^{-19}$ C	$m_p = 1.67 \times 10^{-27}$ kg	$m_e = 9.1 \times 10^{-31}$ kg	$\epsilon_o = 8.85 \times 10^{-12}$ C^2/N·m^2
$\mu_o = 4\pi \times 10^{-7}$ T · m/A	1 eV $= 1.6 \times 10^{-19}$ J	$c = 3.0 \times 10^8$ m/s	$360^\circ = 2\pi$ rad
milli (m) $= 10^{-3}$	micro (μ) $= 10^{-6}$	nano (n) $= 10^{-9}$	pico (p) $= 10^{-12}$
kilo (k) $= 10^3$	Mega (M) $= 10^6$	Giga (G) $= 10^9$	Tera (T) $= 10^{12}$
$k = 1/(4\pi\epsilon_o) = 9 \times 10^9$ N·m^2/C^2		Sphere: $A = 4\pi R^2$ $V = \frac{4\pi}{3}R^3$	

1. In the figure, particles 1 and 3 are fixed in place, but particle 2 is free to move. If the net electrostatic force on particle 2 due to particles 1 and 3 is zero and $L_{23} = 4 \times L_{12}$, what is the ratio q_3/q_1?

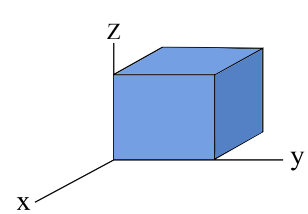


(1) 16 (2) 2 (3) 1 (4) 8 (5) 4

2. An electric dipole with dipole moment $\vec{p} = \left(3\hat{i} + 4\hat{j}\right) 1.24 \times 10^{-30}$ Cm is in an electric field $\vec{E} = 4000$ N/C$\hat{z}$. What is the magnitude of the torque acting on the electric dipole?

(1) 2.48×10^{-26} Nm (2) 1.98×10^{-26} Nm (3) 1.49×10^{-26} Nm (4) 3.47×10^{-25} Nm (5) 0

3. The figure shows a closed Gaussian surface in the shape of a cube of edge length 3.0 m. It lies in a region where the non uniform electric field, measured in N/C, is $\vec{E} = (x + 1)\hat{i} + 2\hat{j} + 4\hat{k}$. What is the net electric charge inside the cube?



(1) 240 pC (2) 80 pC (3) 140 pC (4) 0 (5) 27 pC

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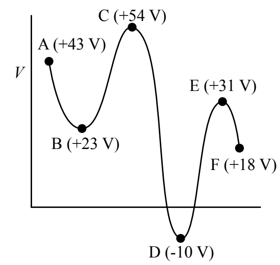
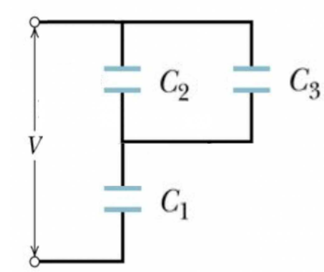
4. The figure shows the electric potential as a function of position. Consider now a particle with charge $-10\ \mu$C moving to the right from position A in the figure. How much kinetic energy does it need to reach F? (Note that the charge is negative!)



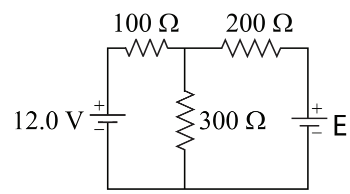
(1) 530 μJ
(2) 250 μJ
(3) 630 μJ
(4) 110 μJ
(5) 310 μJ

5. In the figure a potential difference of $V = 100$ V is applied across a capacitor arrangement with capacitances $C_1 = 10\ \mu$F, $C_2 = 6\ \mu$F and $C_3 = 4\ \mu$F. What is the stored energy U_3 for capacitor 3?



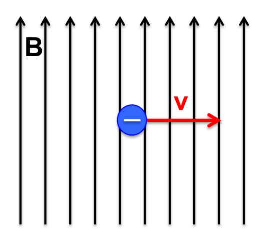
(1) 5.0 mJ
(2) 12.5 mJ
(3) 7.5 mJ
(4) 20.0 mJ
(5) 25.0 mJ

6. In the circuit shown, find the emf $\mathcal{E}$ that makes the current through the 100 Ω resistor zero.



(1) 20 V
(2) 12 V
(3) 9.0 V
(4) 16 V
(5) 24 V

7. An electron moves to the right in magnetic field B pointing up, as shown in the figure. Where does the force experienced by the electron points to?

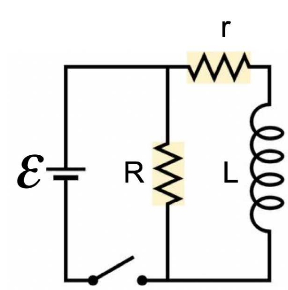


(1) into the page
(2) up
(3) out of page
(4) down
(5) force is zero

8. A proton traveling at 40 m/s in the $+x$ direction enters a region of space in which there is a uniform electric field of 20 N/C pointing in the $-z$ direction. Which of the following magnetic fields (in tesla) would ensure that the proton continues to move in the $+x$ direction without any deflections?

(1) 0.5, $+y$ direction (2) 0.5, $-y$ direction (3) 2.0, $+y$ direction (4) 2.0, $-y$ direction (5) 2.0, $-x$ direction

9. The switch in the circuit shown in the figure has been closed for a long time. What is the difference of potentials across resistor R at the first instance after someone opens the switch? Given that emf $\mathcal{E} = 6$ V, L = 1 H, R = 2 Ω and r = 1 Ω.



(1) 12 V
(2) 24 V
(3) 6 V
(4) 8 V
(5) 4 V

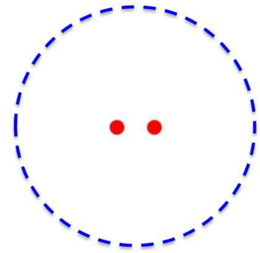
10. An LC circuit has a capacitance of 30 μF and inductance of 60 mH. At time $t = 0$ the charge on the capacitor is 50 μC and the current is 50 mA. The maximum charge on the capacitor is:

(1) 84 μC (2) 96 μC (3) 72 μC (4) 66 μC (5) 58 μC

11. An inverted image is formed 80 cm from an object by a thin lens located between the two. The image is 1/4 the height of the object. What is the distance from the object to the lens?

(1) 64 cm (2) 48 cm (3) 24 cm (4) 32 cm (5) 16 cm

12. Two coherent radiowave point sources separated by $d = 2.5$ m radiate in phase at frequency $f = 400$ MHz. A radiowave detector moves in a circular path of some radius $R > d$ around the mid-point between the two sources (see figure) and measures the radiowave intensity of the signal. Find how many maxima it detects.

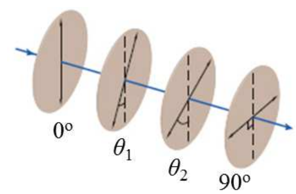


(1) 14
(2) 8
(3) 10
(4) 2
(5) 6

13. The electric field of a radio wave in vacuum has amplitude 2.7×10^{-2} V/m and wavelength of 0.6 μm. What is the amplitude of the magnetic field of this radio wave, in T?

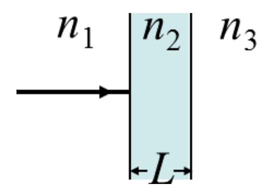
(1) 9.0×10^{-11} (2) 5.4×10^{-11} (3) 1.8×10^{-11} (4) 2.7×10^{-11} (5) 2.1×10^{-11}

14. Unpolarized light is incident on four polarizing sheets with their transmission axes oriented as shown in the figure, where $\theta_1 = 25°$ and $\theta_2 = 45°$. All angles are between the transmission axis and the vertical, and are not necessarily drawn to scale. What percentage of the initial light intensity is transmitted through this set of polarizers?



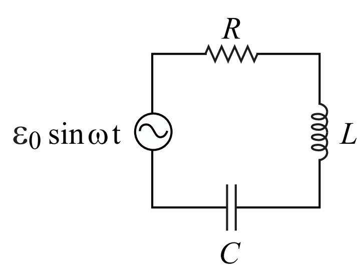
(1) 18.1% (2) 36.2% (3) 30.1% (4) 10.3% (5) 20.5%

15. In the figure, light of frequency $f = 6 \times 10^{14}$ Hz is incident perpendicularly on a thin layer of material 2 that lies between (thicker) materials 1 and 3. The indices of refraction of the materials are $n_1 = 1.6$, $n_2 = 2.5$, and $n_3 = 2.0$. What is the minimum thickness L (in nm) of the thin layer that minimizes transmission of the light?



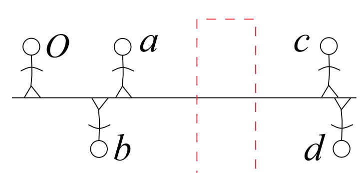
(1) 50 (2) 100 (3) 125 (4) 150 (5) 200

16. Consider the RLC circuit shown in the figure on the right with $R = 10\,\Omega$, $L = 1\,$mH, $C = 5\,\mu$F, and an alternating emf source of $\mathcal{E} = \mathcal{E}_0 \sin(\omega t)$. At what frequency f is the current through the resistor maximal?



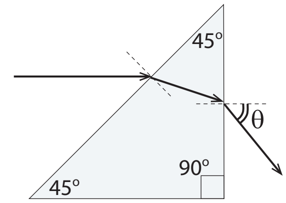
(1) 2.25 kHz (2) 14.1 kHz (3) 4.50 kHz (4) 7.07 kHz (5) 28.3 kHz

17. As shown, stick figure O stands in front of a thin lens that is mounted within the boxed region. The solid horizontal line indicates the central axis of the lens. The four stick figures a to d suggest *general locations and orientations* for the images that might be produced by the lens. (Neither their height nor their distance from the lens is drawn to scale.) Which of the stick figures could not possibly represent images?



(1) b and c (2) b and d (3) a and d (4) a and c (5) only d

18. A ray of light traveling horizontally enters a prism as shown in the figure. The index of refraction of the prism is $n = 1.4$, and the prism is surrounded by air. What is the value of the angle θ between the horizontal and the direction of the ray leaving the prism?



(1) 21° (2) 31° (3) 69° (4) 59° (5) 49°

19. The Thirty Meter Telescope will be commissioned in year 2027. Its reflective mirror will be 30 m in diameter, as the telescope name suggests. In December 2027, Mars will be at its closest approach to Earth of about 80,000,000 km (such close approaches happen approximately once in three years). Approximately, what is the smallest size of an object that this telescope will be able to resolve on Mars at that time? The wavelength of visible light ranges from 400 to 600 nm.

(1) 2 km (2) 200 m (3) 20 m (4) 2 cm (5) 20 cm

20. A small rocket of mass 100 kg emits an intense light beam as a rocket exhaust. What must be the minimum total power (intensity times area) of the beam (in gigawatts) needed to lift the rocket against the force of gravity at the earth's surface ($g = 9.8$ m/s^2)?

(1) 300 (2) 150 (3) 600 (4) 450 (5) 900